

Flat Modules and Tor: Solutions

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Math 121

July 26, 2026

Problem 1. (*Proving Proposition 3*) Let R be a ring and M an R -module.

- a) Let $I \subset R$ be an ideal. Prove that $I \otimes_R M \rightarrow M$ is an injection iff $\mathrm{Tor}_1^R(R/I, M) = 0$ by considering the long exact sequence of Tor derived from the short exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$. Then show that when M is flat, this condition is satisfied for all finitely generated ideals $I \subset R$.
- b) Suppose that the injectivity condition holds for all finitely generated ideals I . Show that for an arbitrary ideal $I' \subset R$ that is not necessarily finitely generated, the multiplication map $I' \otimes_R M \rightarrow M$ is injective.
- c) Let $N' \subset N$ be an inclusion of R -modules. To prove the injectivity of $N' \otimes_R M \rightarrow N \otimes_R M$, show that it suffices to consider the case where N is finitely generated and N/N' is cyclic.
- d) Recall from the modules worksheet that every cyclic R -module is isomorphic to R/I for some ideal $I \subset R$. Assuming $N/N' \cong R/I$, use the long exact sequence of Tor derived from $0 \rightarrow N' \rightarrow N \rightarrow N/N' \rightarrow 0$ to show that $N' \otimes_R M \rightarrow N \otimes_R M$ is injective. Conclude that M is flat and explain why we have just shown that M is flat iff the multiplication map $I \otimes_R M \rightarrow M$ is injective for all finitely generated ideals $I \subset R$.

Proof.

- a) We know that the long exact sequence of Tor derived from the short exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ contains the sequence

$$\mathrm{Tor}_1^R(R, M) \longrightarrow \mathrm{Tor}_1^R(R/I, M) \longrightarrow I \otimes_R M \longrightarrow R \otimes_R M.$$

We already know that

$$\mathrm{Tor}_1^R(R, M) = 0$$

since R is free over itself, and we may also identify $R \otimes_R M \cong M$. Hence, the sequence becomes

$$0 \longrightarrow \mathrm{Tor}_1^R(R/I, M) \longrightarrow I \otimes_R M \xrightarrow{r \otimes m \mapsto rm} M.$$

We now see that the multiplication map $I \otimes_R M \rightarrow M$ is an injection iff $\text{Tor}_1^R(R/I, M) = 0$ by the exactness of the sequence.

Now, when M is flat, we know that

$$\text{Tor}_1^R(N, M) = 0$$

for all R -modules N by Proposition 2, so it surely must hold for $N = R/I$ for any finitely generated ideal $I \subset R$.

- b) Let $I' \subset R$ be an arbitrary ideal. Let $x \in I' \otimes_R M$ map to $0 \in M$ under the multiplication map. We shall show that $x = 0$ in $I' \otimes_R M$ to get injectivity. Since elements of $I' \otimes_R M$ are finite sums of elementary tensors, we can write

$$x = \sum_{j=1}^t r_j \otimes m_j$$

for $r_j \in I'$ and $m_j \in M$. Now, define $I := \langle r_1, \dots, r_t \rangle \subset I'$. Then I is finitely generated. Furthermore, if

$$\tilde{x} = \sum_{j=1}^t r_j \otimes m_j \in I \otimes_R M,$$

then $\tilde{x} \mapsto x$ under the natural map $I \otimes_R M \rightarrow I' \otimes_R M$. Since x maps to $0 \in M$, it follows that $\tilde{x}$ maps to $0 \in M$ under the multiplication map $I \otimes_R M \rightarrow M$. By hypothesis, this map is injective since I is finitely generated. Hence, $\tilde{x} = 0$, and therefore its image $x \in I' \otimes_R M$ is also zero by factoring through $I' \otimes_R M$. Hence, the multiplication map $I' \otimes_R M \rightarrow M$ has trivial kernel and is injective.

- c) Suppose that $x \in N' \otimes_R M$ maps to $0 \in N \otimes_R M$ by the map induced by the functor $- \otimes M$ on the inclusion $N' \hookrightarrow N$. Since x is a finite sum of elementary tensors, the equality $x = 0$ in $N \otimes_R M$ is obtained by using finitely many applications of the defining bilinear relations of the tensor product. Hence, only finitely many elements of N are involved in this equality. Thus, we may pass to a finitely generated submodule $N_0 \subset N$ generated by these elements, and consider the injectivity of the map

$$(N' \cap N_0) \otimes_R M \rightarrow N_0 \otimes_R M.$$

Hence, it suffices to consider the case where N is finitely generated.

To reduce to the cyclic case, notice that when N is finitely generated, the quotient N/N' is also finitely generated. Now, choose a finite set of generators $\bar{x}_1, \dots, \bar{x}_n$ of N/N' , and lift them to elements $x_1, \dots, x_n \in N$ by choosing representatives. Then, let us define

$$N_i = N' + Rx_1 + \dots + Rx_i,$$

so that $N_0 = N'$ and $N_n = N$. This creates a chain of modules

$$N' = N_0 \subset N_1 \subset \dots \subset N_n = N,$$

and by construction, we know that N_{i+1}/N_i is generated by the single class of x_{i+1} since N_{i+1} is obtained from N_i by adding the generator x_{i+1} . Hence, instead of proving that $N' \otimes_R M \rightarrow N \otimes_R M$ is injective at once, it suffices to prove that each $N_i \otimes_R M \rightarrow N_{i+1} \otimes_R M$ is injective. Thus, it suffices to prove injectivity in the case where N/N' is cyclic (and finitely generated).

- d) Assume N/N' is cyclic and finitely generated. Then we may write $N/N' \cong R/I$ for some ideal $I \subset R$. The short exact sequence $0 \rightarrow N' \rightarrow N \rightarrow N/N' \rightarrow 0$ induces the long exact sequence of Tor, which contains the sequence

$$\mathrm{Tor}_1^R(N/N', M) \longrightarrow N' \otimes_R M \longrightarrow N \otimes_R M.$$

Since we have $N/N' \cong R/I$, we may write

$$\mathrm{Tor}_1^R(N/N', M) \cong \mathrm{Tor}_1^R(R/I, M),$$

but by part (a) and (b), this vanishes since the multiplication map $I \otimes_R M \rightarrow M$ is injective for arbitrary ideals $I \subset R$. Hence, the map $N' \otimes_R M \rightarrow N \otimes_R M$ is injective by exactness, and $- \otimes_R M$ must preserve injections. But $- \otimes_R M$ is already right exact, so $- \otimes_R M$ is exact and M is flat.

In part (a), we saw that when M is flat, the multiplication map $I \otimes_R M \rightarrow M$ is injective for all finitely generated ideals $I \subset R$. Conversely, in part (b), (c), and (d), we assumed that the multiplication map $I \otimes_R M \rightarrow M$ was injective for all finitely generated ideals $I \subset R$, then showed that $- \otimes_R M$ preserves injections $N' \hookrightarrow N$ by first extending the injectivity condition to all ideals of R , then reducing to the case where N was finitely generated and N/N' was cyclic. This showed that M was flat, hence completing the proof of Proposition 3.

□

Problem 2. (*Corollary 4 and Applications*) Let $(R, \mathfrak{m})$ be a local Noetherian ring, and let M be a finitely generated R -module.

- a) Prove that if M is flat, then M is free. (Hint: Choose elements $m_1, \dots, m_t \in M$ whose images form a basis of the vector space $M/\mathfrak{m}M$ over $R/\mathfrak{m}$. Then use Nakayama to obtain a surjection $R^t \rightarrow M$, and use flatness to force the kernel to vanish)
- b) Take $R = \mathbb{K}[x]_{\langle x \rangle}$ with the unique maximal ideal $\mathfrak{m} = \langle x \rangle$, and consider

$$M = R/\langle x \rangle \cong \mathbb{K}.$$

Prove that M is not flat over R by using the free resolution

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

to calculate $\text{Tor}_1^R(R/\mathfrak{m}, M)$.

- c) Let $R = \mathbb{Z}_{\langle p \rangle}$, the localization of $\mathbb{Z}$ at the prime ideal $\langle p \rangle$, and consider

$$M = R/\langle p \rangle \cong \mathbb{F}_p.$$

Show that M is not flat over R by calculating $\text{Tor}_1^R(R/\langle p \rangle, M)$.

Proof.

- a) Since M is a finitely generated R -module and $(R, \mathfrak{m})$ is a local ring, we know that $M/\mathfrak{m}M$ is a finite-dimensional vector space over the residue field $R/\mathfrak{m}$. Hence, we can choose elements $m_1, \dots, m_t \in M$ whose images in the quotient form a basis of $M/\mathfrak{m}M$. By (Local) Nakayama's Lemma III, these elements in fact form a generating set for M . Hence, we have a surjection $\varphi : R^t \rightarrow M$ given by $e_i \mapsto m_i$. We want to prove that φ is an isomorphism to show that M is free.

Let us set $K = \ker \varphi$. Then, we have the short exact sequence

$$0 \longrightarrow K \longrightarrow R^t \longrightarrow M \longrightarrow 0.$$

Now, by the M flat iff $\text{Tor}_1^R(R/\mathfrak{m}, M) = 0$ part of Corollary 4, the long exact sequence of Tor given by tensoring the short exact sequence above with $R/\mathfrak{m}$ contains

$$0 = \text{Tor}_1^R(R/\mathfrak{m}, M) \longrightarrow K \otimes_R R/\mathfrak{m} \longrightarrow R^t \otimes_R R/\mathfrak{m} \longrightarrow M \otimes_R R/\mathfrak{m} \longrightarrow 0,$$

which is another short exact sequence. Calculating the tensor products, we get the short exact sequence

$$0 \longrightarrow K/\mathfrak{m}K \longrightarrow (R/\mathfrak{m})^t \longrightarrow M/\mathfrak{m}M \longrightarrow 0,$$

but notice that the map $(R/\mathfrak{m})^t \rightarrow M/\mathfrak{m}M$ is an isomorphism by our choice of $m_1, \dots, m_t$. Thus,

$$0 = \ker((R/\mathfrak{m})^t \rightarrow M/\mathfrak{m}M) = \text{im}(K/\mathfrak{m}K \rightarrow (R/\mathfrak{m})^t) \cong K/\mathfrak{m}K$$

by exactness, and we get that $K/\mathfrak{m}K \cong 0$, i.e. $\mathfrak{m}K = K$. Since $K \subset R^t$ is a submodule of a finitely generated module over a Noetherian ring, K is finitely generated. By (Local) Nakayama's Lemma II, we know that $K = 0$, so φ is in fact injective, and hence an isomorphism of R -modules. Thus, $R^t \cong M$ and M is free.

b) We tensor the free resolution for $R/\mathfrak{m}$ in the problem statement by M to see that

$$0 \longrightarrow M \xrightarrow{\cdot x} M \longrightarrow 0,$$

so that

$$\mathrm{Tor}_1^R(R/\mathfrak{m}, M) \cong \frac{\ker(\cdot x)}{\mathrm{im} 0} \cong \ker(\cdot x).$$

But notice that $M = R/\langle x \rangle$, so x acts as 0 on M , and hence $M \xrightarrow{\cdot x} M$ is the zero map. Hence,

$$\mathrm{Tor}_1^R(R/\mathfrak{m}, M) \cong M \neq 0,$$

and by Corollary 4, M is not flat over R .

c) The proof of this is actually entirely the same as part (b). Indeed, tensoring the free resolution

$$0 \longrightarrow R \xrightarrow{\cdot p} R \longrightarrow R/\langle p \rangle \longrightarrow 0$$

for $R/\langle p \rangle$ by M , we get

$$0 \longrightarrow \mathbb{F}_p \xrightarrow{\cdot p} \mathbb{F}_p \longrightarrow 0$$

and the multiplication by p map clearly acts as the zero map in $\mathbb{F}_p$, so that

$$\mathrm{Tor}_1^R(R/\langle p \rangle, \mathbb{F}_p) \cong \ker(\cdot p) \cong \mathbb{F}_p.$$

Thus, $\mathbb{F}_p$ is not flat as a $\mathbb{Z}_{\langle p \rangle}$ -module by Corollary 4.

□

Problem 3. (*Computations with Tor*) Let $R = \mathbb{Z}$.

a) Use the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow 0$$

for $\mathbb{Z}/n\mathbb{Z}$ to compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ for any $\mathbb{Z}$ -module (abelian group) A . What happens if A is flat? Compare with Corollary 2.

b) Compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$.

c) Let R be a ring and suppose that $x \in R$ is a nonzerodivisor, with M an R -module. Calculate $\mathrm{Tor}_i^R(R/\langle x \rangle, M)$ by using the fact that

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

is a free resolution of $R/\langle x \rangle$. Compare with parts (a) and (b).

Proof.

a) Tensoring the free resolution in the problem statement with A , we get

$$0 \longrightarrow A \xrightarrow{\cdot n} A \longrightarrow 0$$

after identifying

$$\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A.$$

Now, we calculate

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong \frac{\ker(\cdot n)}{\mathrm{im} 0} \cong \{a \in A : na = 0\} = (0 :_A n).$$

When A is flat, we know that by Proposition 2, we have

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = (0 :_A n) = 0,$$

so that in particular, A is torsion free. This matches what we expect from Corollary 2 since $\mathbb{Z}$ is a PID, and we just showed that A being flat implies A is torsion free.

b) In the case where $A = \mathbb{Z}/m\mathbb{Z}$, we have

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) = \{a \in \mathbb{Z}/m\mathbb{Z} : na = 0\},$$

but we know that the elements a that are killed by n in $\mathbb{Z}/m\mathbb{Z}$ are those such that $m \mid na$, which are exactly

$$\overline{0}, \overline{\frac{m}{d}}, \overline{2\frac{m}{d}}, \dots, \overline{(d-1)\frac{m}{d}},$$

where $d = \gcd(n, m)$, so that

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/d\mathbb{Z}.$$

c) Tensoring the free resolution given in the problem statement by M , we get

$$0 \longrightarrow M \xrightarrow{\cdot x} M \longrightarrow 0 ,$$

so that the only nontrivial Tor modules are $\text{Tor}_i(R/\langle x \rangle, M)$ for $i = 0, 1$, since the sequence is zero for $i > 1$. Indeed, we know that

$$\text{Tor}_0^R(R/\langle x \rangle, M) = R/\langle x \rangle \otimes_R M \cong M/xM$$

by Example 2, and we see that

$$\text{Tor}_1^R(R/\langle x \rangle, M) \cong \frac{\ker \cdot x}{\text{im } 0} \cong \{m \in M : xm = 0\} = (0 :_M x),$$

so that we obtain

$$\text{Tor}_i^R(R/\langle x \rangle, M) \cong \begin{cases} M/xM, & i = 0, \\ (0 :_M x), & i = 1, \\ 0, & i > 1. \end{cases}$$

□

Problem 4. (*Applications of Corollary 2*) Let $R = \mathbb{Z}$. Which one of the following $\mathbb{Z}$ -modules are flat?

$$\mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{Z}/n\mathbb{Z}, \quad \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}, \quad \mathbb{Q}/\mathbb{Z}, \quad \mathbb{Z}_{\langle p \rangle}.$$

Proof. Note that $\mathbb{Z}$ and $\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ are immediately free $\mathbb{Z}$ -modules, so they must be flat.

For the rest, we may apply Corollary 2, which says that if R is a PID (which $\mathbb{Z}$ is), then M is flat iff M is torsion free, i.e. for every nonzero $r \in R$, we have $rm = 0 \implies m = 0$.

Using this result, notice that $\mathbb{Q}$ is torsion-free as a $\mathbb{Z}$ -module, since if

$$nq = 0$$

for some nonzero $n \in \mathbb{Z}$, then $q = 0$. Hence, $\mathbb{Q}$ is flat as a $\mathbb{Z}$ -module.

On the other hand, $\mathbb{Z}/n\mathbb{Z}$ is not torsion free as a $\mathbb{Z}$ -module for $n \geq 2$, since

$$n\bar{1} = \bar{0}$$

but $\bar{1} \neq \bar{0}$. The only exception is when $n = 1$ and $\bar{1} = \bar{0}$. Hence, $\mathbb{Z}/n\mathbb{Z}$ is a non-flat $\mathbb{Z}$ -module for $n \geq 2$.

Likewise, $\mathbb{Q}/\mathbb{Z}$ is not torsion free since

$$\frac{1}{2} + \mathbb{Z} \neq 0,$$

but

$$2 \left(\frac{1}{2} + \mathbb{Z} \right) = 1 + \mathbb{Z} = \bar{0},$$

so that $\mathbb{Q}/\mathbb{Z}$ is not flat.

Finally, we may view $\mathbb{Z}_{\langle p \rangle}$ as a subgroup of $\mathbb{Q}$, and $\mathbb{Q}$ is torsion free, so that $\mathbb{Z}_{\langle p \rangle}$ is also torsion free and hence flat by Corollary 2. $\square$

Problem 5. (*Applications of Corollary 3*) Consider the local Noetherian rings $(R, \mathfrak{m}) = (\mathbb{K}[x]_{\langle x \rangle}, \langle x \rangle)$ and $(S, \mathfrak{n}) = (\mathbb{K}[[x]], \langle x \rangle)$.

- a) Explain how we can think about S as an R -algebra. Also show that $\mathfrak{m}S \subset \mathfrak{n}$.
- b) Use the free resolution

$$0 \longrightarrow R \xrightarrow{\cdot x} R \longrightarrow R/\langle x \rangle \longrightarrow 0$$

to show that

$$\mathrm{Tor}_1^R(R/\langle x \rangle, S) \cong \ker(S \xrightarrow{\cdot x} S).$$

Then show that this kernel is 0, and deduce that $\mathbb{K}[[x]]$ is flat as a $\mathbb{K}[x]_{\langle x \rangle}$ -module.

Proof.

- a) First, note that there is a natural inclusion $\varphi : \mathbb{K}[x] \hookrightarrow \mathbb{K}[[x]]$ given by sending a polynomial to the same expression regarded as a formal power series. Note that $\varphi(\mathbb{K}[x] \setminus \langle x \rangle)$ consists of power series $f(x)$ with $f(0) \neq 0$, which are hence invertible in $\mathbb{K}[[x]]$. By the universal property of localization, we obtain a map $\tilde{\varphi} : \mathbb{K}[x]_{\langle x \rangle} \rightarrow \mathbb{K}[[x]]$, which gives $\mathbb{K}[[x]]$ a $\mathbb{K}[x]_{\langle x \rangle}$ -algebra structure. Notice that the extension

$$\mathfrak{m}S = \langle x \rangle S = \langle x \rangle = \mathfrak{n},$$

so that in particular, we have $\mathfrak{m}S \subset \mathfrak{n}$.

- b) Tensoring the free resolution for $R/\langle x \rangle$ in the problem statement with S , we get

$$0 \longrightarrow S \xrightarrow{\cdot x} S \longrightarrow 0$$

where we make the identification $R \otimes_R S \cong S$. Now, we see that

$$\mathrm{Tor}_1^R(R/\langle x \rangle, S) \cong \frac{\ker(\cdot x)}{\mathrm{im} 0} \cong \ker(S \xrightarrow{\cdot x} S).$$

To see that this kernel vanishes, take some power series

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots \in S$$

and suppose that $xf(x) = 0$. Then

$$xf(x) = a_0x + a_1x^2 + a_2x^3 + \cdots = 0.$$

Matching coefficients, we get

$$a_0 = a_1 = a_2 = \cdots = 0,$$

so that $f(x) \equiv 0 \in S$. Hence, the kernel is trivial, and by Corollary 3, we know that $\mathbb{K}[[x]]$ is flat as a $\mathbb{K}[x]_{\langle x \rangle}$ -module since $\mathrm{Tor}_1^R(R/\mathfrak{m}, S) = 0$.

□